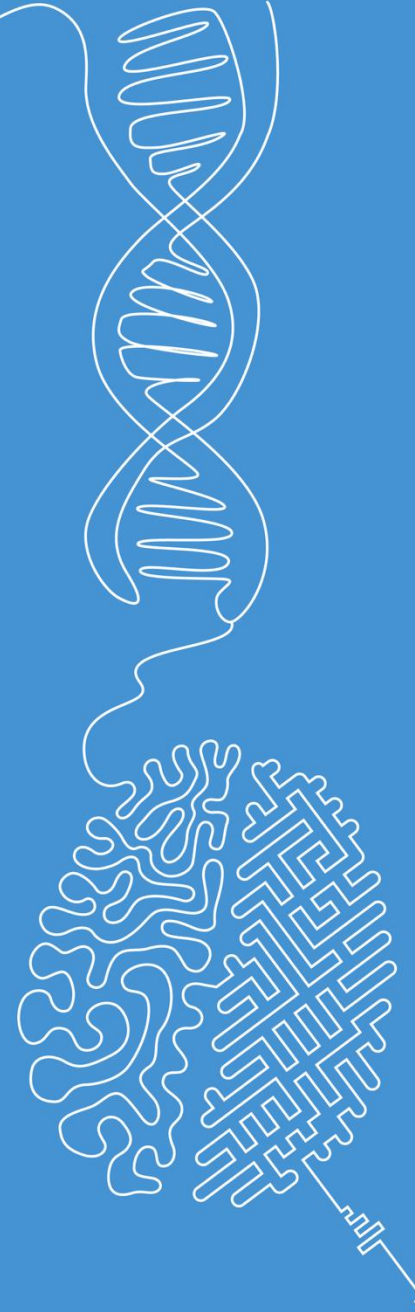


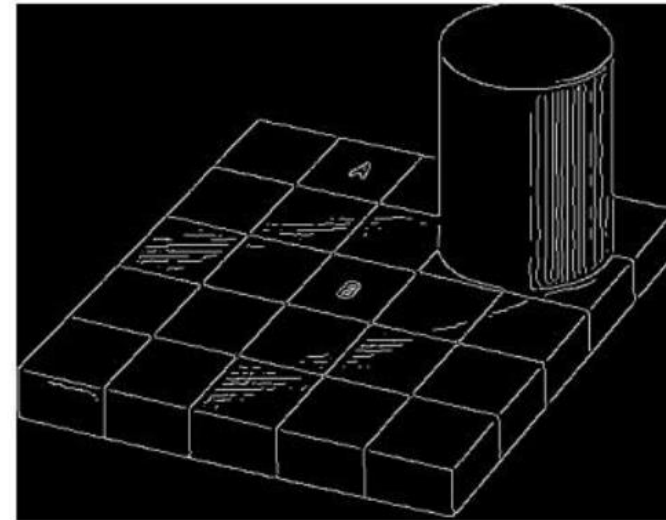
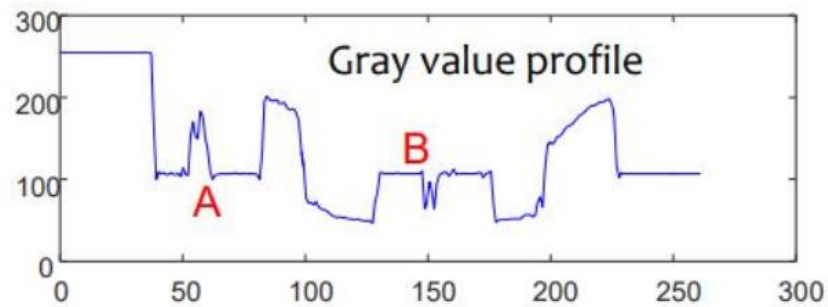
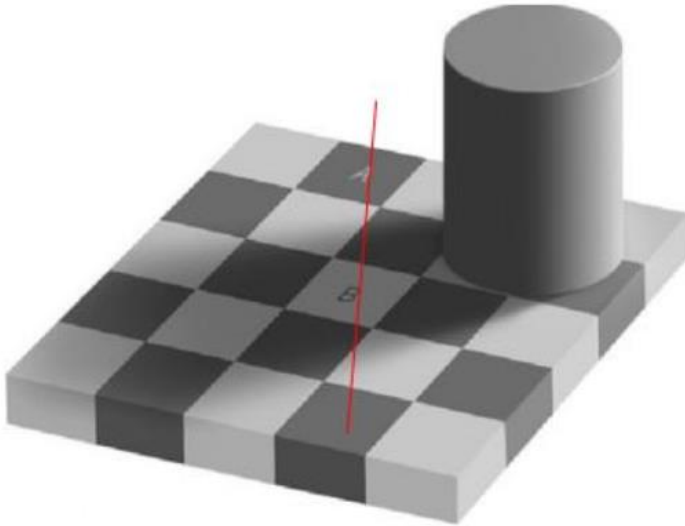
Edge detection

Image and Signal Processing

Norman Juchler



Edge detection



Edges contain essential, local image information. Consequently, the detection of edges with edge filters is a fundamental operation in image processing.

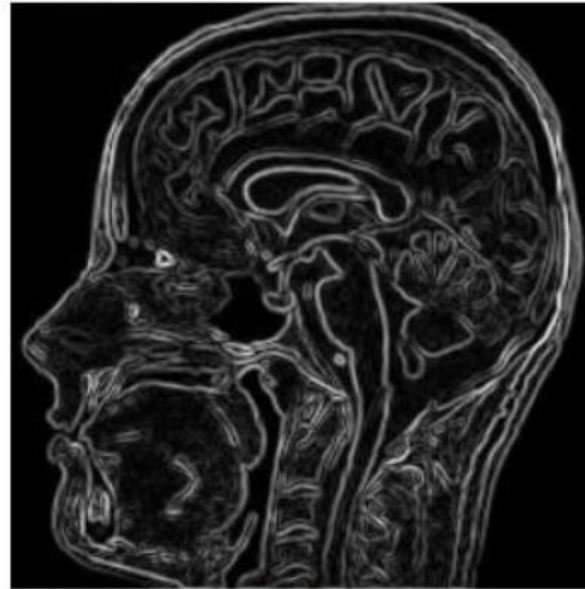
Edge detection: Methods

The focus can be set on edge strength or edge localization or edge direction



original

edge strength



gradient magnitude

edge position



canny

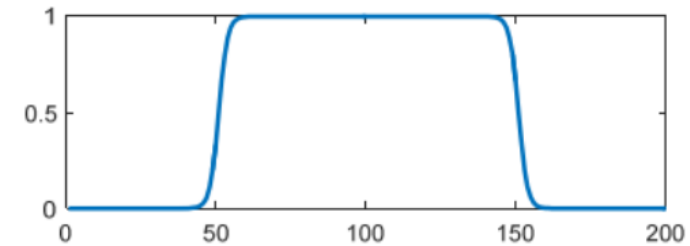
- directed edge filter, gradient magnitude
- Laplace filter, LoG filter, DoG filter
- Canny filter

Edge detection: Concept of image derivatives (1D)

Signal: Image

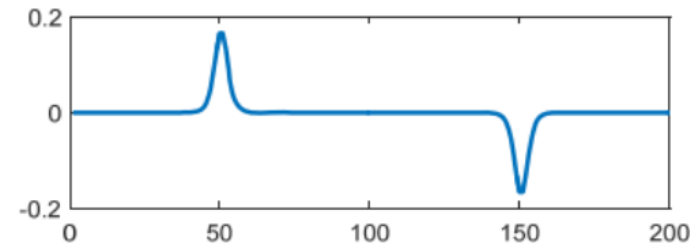


$$f(x, y)$$



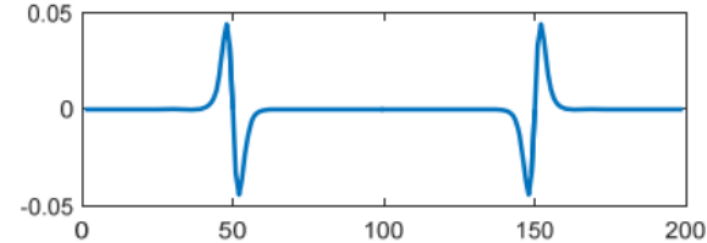
first Derivative

$$\frac{\partial}{\partial x} f(x, y)$$



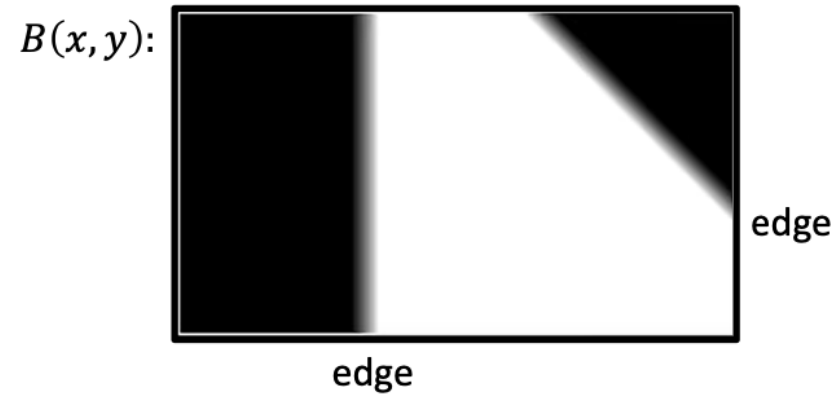
second Derivative

$$\frac{\partial^2}{\partial x^2} f(x, y)$$



Edge detection: Concept of image derivatives (2D)

2-Dimensions



The partial derivative in 2-dimensions represents the amount of change /gradient in directions x and y in an image.

$$\nabla B = \left[\frac{\partial B}{\partial x}, \frac{\partial B}{\partial y} \right] \quad \text{Gradient- or „Del“- Operator}$$

Recommendation: <https://www.youtube.com/watch?v=IOEBsQodtEQ&t=148s>

Edge detection: Concept of image derivatives (2D)

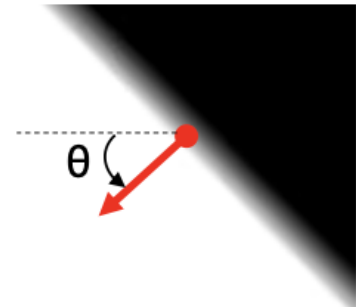
2-Dimensions

Gradient Magnitude

$$S = \|\nabla B\| = \sqrt{\left(\frac{\partial B}{\partial x}\right)^2 + \left(\frac{\partial B}{\partial y}\right)^2}$$

Gradient Orientation

$$\theta = \tan^{-1} \left(\frac{\frac{\partial B}{\partial y}}{\frac{\partial B}{\partial x}} \right)$$



Edge detection: Concept of image derivatives (2D)

2-Dimensions

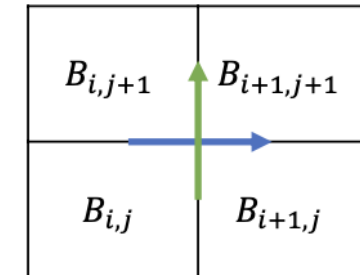
Finite-difference approximation to get formulas for discrete images

$$\frac{\partial B}{\partial x} \approx \left((B_{i+1,j+1} - B_{i,j+1}) + (B_{i+1,j} - B_{i,j}) \right)$$

$$\frac{\partial B}{\partial y} \approx \left((B_{i+1,j+1} - B_{i+1,j}) + (B_{i,j+1} - B_{i,j}) \right)$$

Change in x-direction

Change in y-direction



image

This can be expressed as **filter masks**:

$$\frac{\partial B}{\partial x} \approx \begin{array}{|c|c|} \hline -1 & 1 \\ \hline -1 & 1 \\ \hline \end{array}$$

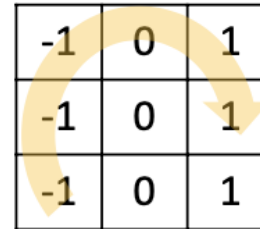
$$\frac{\partial B}{\partial y} \approx \begin{array}{|c|c|} \hline 1 & 1 \\ \hline -1 & -1 \\ \hline \end{array}$$

Edge detection: Concept of image derivatives (2D)

2-Dimensions

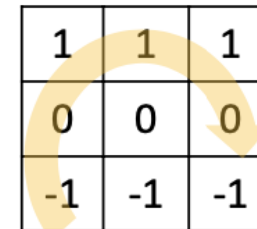
3x3 Prewittfilter

x-direction



-1	0	1
-1	0	1
-1	0	1

y-direction



1	1	1
0	0	0
-1	-1	-1

Diagonal directions

-1	-1	0
-1	0	1
0	1	1

0	1	1
-1	0	1
-1	-1	0

The increase in filter size allows for **regularization** (noise will be suppressed by taking into account a larger image area).

Edge detection: Concept of image derivatives (2D)

2-Dimensions

3x3 Sobelfilter

x-direction

-1	0	1
-2	0	2
-1	0	1

y-direction

1	2	1
0	0	0
-1	-2	-1

To further enhance the regularization, the mask elements can be weighted.

Edge filter masks can be constructed for any dimension:

- **Small filter masks** → good localization, sensitive for noise
- **Large filter masks** → bad localization, robust against noise

Edge detection: Concept of image derivatives (2D)

A last step of the gradient method, via a threshold it is decided which pixels belong to an edge and which don't.

Simple threshold T:

$$\begin{aligned}\|\nabla B(x, y)\| < T & \quad \text{belongs to edge} \\ \|\nabla B(x, y)\| \geq T & \quad \text{does not belong to edge}\end{aligned}$$

Hysteresis based threshold $T_0 < T_1$:

$$\begin{aligned}\|\nabla B(x, y)\| < T_0 & \quad \text{does not belong to edge} \\ \|\nabla B(x, y)\| \geq T_1 & \quad \text{belongs to edge} \\ T_0 \leq \|\nabla B(x, y)\| < T_1 & \quad \text{belongs to edge, if neighbor pixel belongs to edge as well}\end{aligned}$$